Assignment -9

Problem Statement
Ques are frequently used in computer programming, and a typical example is the creation of a job queue by an operating system. If the operating system does not use priorities then the jobs are processed in the order they enter the system. Write a C program for simulating a job queue. Write functions to add a job and delete a job.

Objective
1. Understand the concept of Queue and FIFO (First In, First Out)
2. Implement a linear queue using an array
3. Implement functions to add and delete jobs.
4. Simulate processing of jobs in the order they enter the system.
5. Analyze the time and space complexity of queue operations.

Implementation
- Platform
  - 64 bit Open source Linux or its derivatives.
  - Open Source C Programming tool like gcc / Eclipse IDE.

- Test Conditions :-

Test Case	Input	Expected Output
TC01	Add one job	Job added successfully
TC02	Add multiple jobs	Jobs stored in FIFO
TC03	Delete a job	First entered Job is del
TC04	Delete from empty queue	Queue underflow
TC05	Add job when queue is full	Queue overflow
TC06	Display pending jobs.	All jobs displayed front to rear

Pseudocode

```
# define MAX 100
int q[MAX], front = -1, rear = -1;

void enqueue (int val){
    if (rear == MAX-1) return print ("Overflow");
    if (front == -1) front = 0;
    q[++rear] = val;
}
void dequeue (){
    if (front == -1 || front > rear) return print ("Underflow");
    print ("Deleted:", q[front++]);
    if (front > rear) front = rear = -1;
}
void displq (){
    if (front == -1) return print ("Empty");
    for (int i = front; i <= rear; i++) print (q[i]);
}
```

Time Complexity

- Enque (Add a job) : O(1)
- Deque (Delete) : O(1)
- Displq Pending Jobs : O(n)
- Overflow / Underflow : O(1)

FAQs

1) What is Queue

Ans A queue is a linear data structure that follows a specific order in which operations are performed: FIFO

2 What is FIFO?

Ans FIFO - First In First Out meaning the element that is added first is the first one to be removed.

3 Why is a queue suitable for an operating system job queue?

Ans It ensures fairness by processing tasks in the exact chronological order of their arrival.

4) What is Queue Overflow and Underflow?

- Overflow: Occurs when trying to add an element to a queue that is already full.
- Underflow: Occurs when trying to delete or remove an element from an empty queue.

5) What are the applications of Queue?

- CPU and disk scheduling in operating systems, print spoolers, handling asynchronous data transfers (buffers, pipes) and Breadth-First Search (BFS) graph traversal.

6) What is a Circular Queue?

- A linear ~~Queue~~ data structure that follows the FIFO principle where the last position is connected back to the first position to form a circle, solving the wasted space issue of linear arrays.

7) Why is a Circular Queue preffered over a Linear Queue?

- In a linear queue, once element are deleted, the spaces at the front cannot be reused even if the rear is at the max capacity. A circular queue efficiently reuses these empty spaces.

8) what principle does a Circular Queue follow?
. The FIFO (First in, First out) principle

9) what are the basic operations performed on a Circular Queue?
- enqueue() (insert), dequeue() (delete), peek() (view front element), and display().

10) What is an overflow & Underflow condition in a Circular Queue?
- Overflow : Occurs when (rear+1) % MAX == front.
- Underflow : Occurs when front == -1.

5/10